Huawei Access Node Optical Transport Failure Resolution SOP for BAS-ACC-07 at Basavanagudi POP

Node: BAS-ACC-07 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_6401

Description

This SOP addresses a Major severity Optical Transport Failure in the Access node BAS-ACC-07 at Basavanagudi POP, running on Huawei equipment. The failure domain is attributed to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. The underlying protocol or hardware mechanics responsible for this failure domain in the Huawei Access node is the Optical Transport Network (OTN) protocol. OTN is used for transporting client signals over optical fibers. It uses lasers or light-emitting diodes (LEDs) for transmitting data and photodetectors for receiving data. The OTN protocol employs a framing structure to ensure reliable and efficient transmission of client signals. This framing structure includes a header and a payload section, where the header contains control information and the payload section carries the client data. In the case of SFP/XFP degradation or fiber cuts, the optical signal may be attenuated or completely lost, leading to a failure in the OTN protocol. Similarly, Rx/Tx power loss can also cause a failure in the OTN protocol. To troubleshoot such failures, the Huawei Access node uses a variety of diagnostic tools and techniques, including the display of alarm information, the display of interface status, and the use of diagnostic commands to isolate the root cause of the failure.

Pre-requisite Checks

- Verify that the Basavanagudi POP is accessible and that remote hands are dispatched to facilitate the troubleshooting process.

- Check the status of the redundant link at the Basavanagudi POP to ensure that it is functioning properly.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display alarm, Expected Output: ALARM_6401: Optical transport failure, Description: This command displays the current alarm information, including the alarm code and a brief description of the alarm. The expected output indicates that the Optical transport
```

```
failure alarm is triggered, which corresponds to the ALARM 6401 code.

Command: display interface, Expected Output: Interface: 1/1/1, State: Down, Description: This command displays the status of the interfaces on the Huawei Access node. The expected output indicates that interface 1/1/1 is down, which may be related to the Optical transport failure.

Command: diag sfp, Expected Output: SFP info: 1/1/1, Status: Degraded, Description: This command executes the diagnostic test on the SFP at interface 1/1/1. The expected output indicates that the SFP is degraded, which is a possible root cause of the Optical transport failure.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace sfp 1/1/1, Description: This command initiates the replacement of the SFP/XFP module at interface 1/1/1. After replacing the SFP/XFP module, verify that the Optical transport failure alarm is cleared.

Step 2: Step: Verify the fiber connections, Command: display fiber, Description: This command displays the status of the fiber connections at the Basavanagudi POP. Verify that the fiber connections are secure and not damaged.

Step 3: Step: Adjust the Rx/Tx power settings, Command: set interface 1/1/1 power-adjustment 1, Description: This command adjusts the Rx/Tx power settings for interface 1/1/1. After adjusting the power settings, verify that the Optical transport failure alarm is cleared.

Post-Check Verification

```
Command: display alarm, Expected Output: No alarm, Description: This command verifies that the Optical transport failure alarm is cleared.

Command: display interface, Expected Output: Interface: 1/1/1, State: Up, Description: This command verifies that interface 1/1/1 is up and functioning properly.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace sfp 1/1/1 restore, Description: This command restores the original SFP/XFP module at interface 1/1/1.

- Step: Revert the fiber connections, Command: display fiber revert, Description: This command reverts the fiber connections to their original state.